Ch-11

Moon : Phases, Facts & Motion

- The day when the moon appears as a full bright circle is called Full moon day (or Purnima. The day when the moon isn't visible in the sky is called New moon day (or Amavasya)

- After the new moon, the period when the bright part of the moon increases is called waxing period. After the full moon, the period when bright portion continues to shrink until ~~is~~ it's no longer visible, is known as waning period. Both of these periods ~~are of~~ last for 2 weeks.

- Waxing period - Shukla Paksh
  Waning period - Krishna Paksh

- The changing of shapes of the moon's bright portion from one day to another as seen from the Earth are called phases of moon.

- On ~~a~~ full moon day, the Moon is nearly opp. to the Sun. As sun rises in the east, the moon sets in west. But, as the bright portion starts to decrease, moon appears to move closer to the sun. When the moon is visible only half, it is overhead at sunrise. Waxing moon is the easiest to spot at sunset while ~~wan~~ waning moon is the easiest to spot ~~on~~ at sunrise.

- Moonrise is seen at different times at different place. Moonrise occurs about 50 minutes later each day. Positional Astronomy Centre (IMD) publishes times for moonrise in your area.

- Sometimes, we see the whole moon but sometimes, we don't see even a bit of it. This is because moon does not have its own light, it just reflects the light of the sun that falls on it. The part on which sunlight fall is ~~called illu~~ becomes illuminated & other portion remains dark.

• When more than half of illuminated portion is visible, it is called Gibbous phase but, when less than half illuminated portion is visible, it is called Cresent phase.

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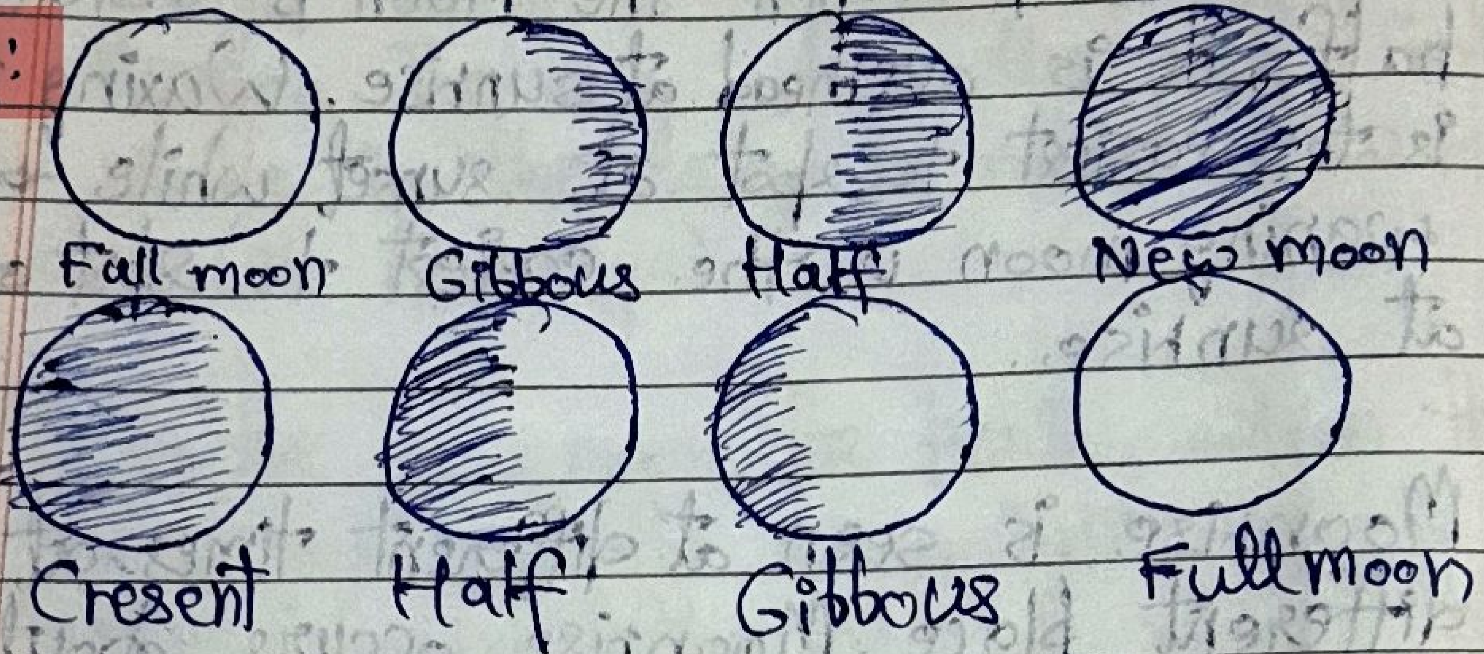
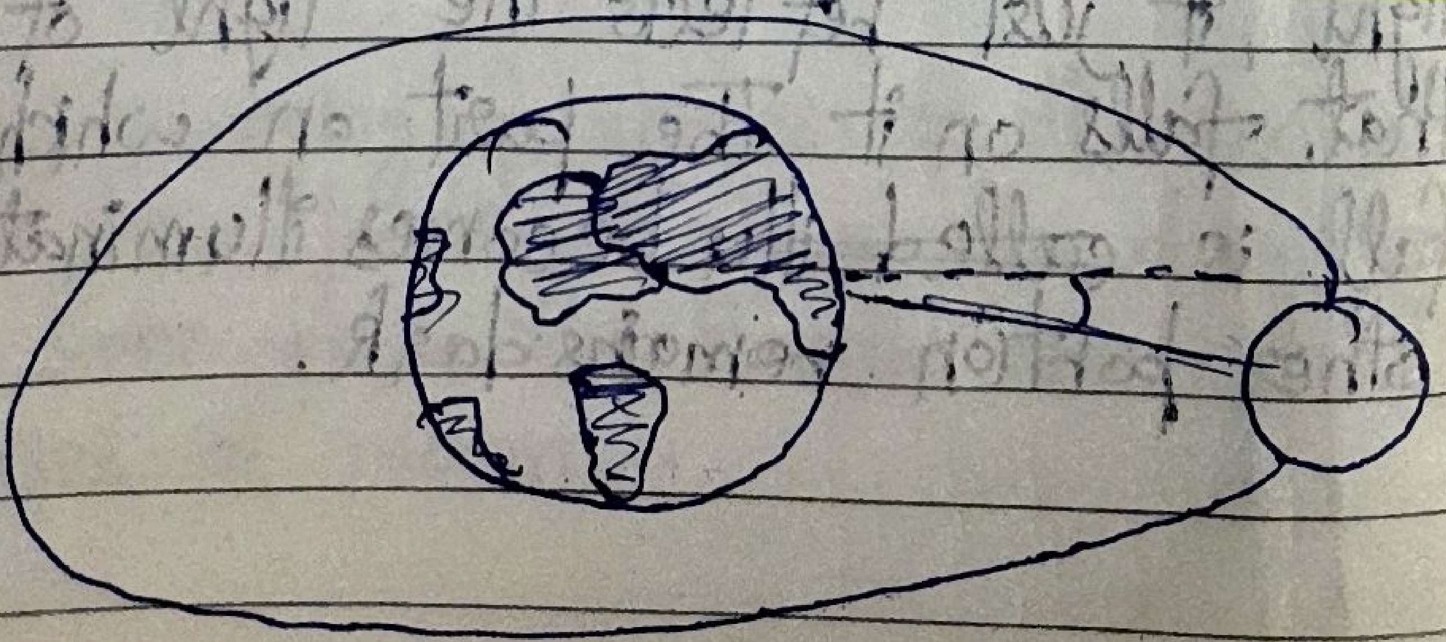


• Lunar eclipses occur only on full moon day & solar eclipses occur only on a new moon day. But, every ~~fu~~ full or new moon day doesn't causes eclipses because the there is a small tilt in moon's orbit with respect to earth's orbit.



Calenders

- The avg. time for Sun to go from the highest point in sky on one day to its highest point in sky the next day, is 24 hours. When shadow is the shortest, the Sun is at the highest point in sky. The period is called mean solar day.

- The cycle of the phases of moon is called a month. In a month, moon' cycle through all it phases & its duration is $29\frac{1}{2}$ days

- Earth take 365 & a quater days to complete 1 revolution around Earth. During this cycle, Earth undergoes 1 whole cycle of season. This period is called solar year.

- In ancient times, people noticed that in a cycle of seasons 12 lunar cycles (cycles of phases of moon) can be fitted. This is how lunar calenders were invented & had 12 months, each of 29.5 days. But, seasons did not stay synchronised in a lunar calender because ~~lunar calender~~ ~~has~~ lunar year has 354 days but a season cycle needs 365 days to repeat.

- For synchronization, people invented solar calender. Knowing seasons was important for agriculture. Gregorian calender is the most used calender & a solar calender. Months are adjusted to fit 365-day cycle. In a solar year, a quater day is left at last. After 4 years, these 4 quaters make a whole day & that day is added to that year to the month of Feb. Such year are called a leap year.

- For a leap year, adding the day is excluded every 100 years but that day is added again every 400 years.

- The time b/w 2 successive spring equinoxes is called tropical year. Gregorian calender is based ~~to~~ on tropical year. The time for the same stars to rise again in sky is called sidereal year. Sidereal year is longer than tropical year by 20 min. Sidereal year is used for keeping track of Earth's pos. in its orbit around the Sun.

- In summers, sun moves a bit north of East & in winter, sun moves a bit south of East. These extremes occur on solistices (Dec 21 & June 21). In Taittiriyā Samhita, this northward movement is called Uttarayan & this southward movement is called Dakshinayan.

- The ~~lunar~~ luni-solar calenders, is a special type of ~~cal~~ calender ~~which is also used as India's main calender~~. Lunar year is just 11 days short compared to Solar year. So, in this calender every 2-3 years, days are added to the year. The days accumulated are nearly a month so a month (Adhik Maasa) is added.

- Indian National Calender is used in India, & has 30 or 31 days & names of the months are taken from ancient Indian months names. Second to sixth months have 31 days & the rest months have 30 days. To match leap year of Gregorian calender, a day is added to Chaitra. In such years, new year begins on 21 March of Gregorian Calender.

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- Festivals are also related to astronomical phenomenons & months. Eg- Eid-ul-fitr after sighting of cresent moon ~~to~~ at the end of Ramadan, Holi in Falghuna. Festivals following luni-solar calender occur ~~only~~ with a ~~d~~ different dates but that shift is less than a month. But, festival following lunar calender can occur in differents months of gregorian calender. Due to wobble in Earth's axis, the festivals following sidereal calender move ahead of of tropical ~~ye~~ calender.

- Man has sent many artificial satelites to the space. Most orbit Earth from 800 KM above & take ~ 100 min to complete 1 orbit. They help in weather monitoring, navigation, communication, etc. ISRO also sent many satellites to the space. Eg- Cartosat (mapping), Mangalyaan (Mars satellite)

- When artificial satellites are sent to space, after their useful life, they become non-functioning metal garbage known as space junk. This can cause danger to working satellites & ~~con~~ countries are working to remove them.

Scientists

1) Vikram Ambalal Sarabhai - Known as the father of Indian space programme. He was a researcher in space science & nuclear physics. VSSC in ~~Th~~ Thiruvanantapuram is an ISRO centre is named after him.

2) Meghnad Saha - Astrophysicist, who studied stars & their temp.. He developed a mathematical equation, known as Saha equation. He was also a member of CRC.